

Category	Information to collect	Case information	Source or note
Case identification	Case name	VELFAC 200 ENERGY - Top-guided window	Official VELFAC EPD and product documentation
Case identification	Source type	Publicly documented manufacturer system	Selected because public product, EPD, guide and warranty information are available
Case identification	Building type	Residential	The system is suitable for domestic projects and multi-plot housing
Case identification	Geographic context	Europe	Manufacturer documentation and EPD relate to European production and market context
Case identification	Window type	Top-guided residential window system	The EPD reference product is a top-guided window
Main system	Frame material	Wood-aluminium composite system with aluminium exterior and timber interior	VELFAC describes the product as wood-aluminium, with aluminium outside and FSC-certified pine inside
Main system	Glazing type	Triple glazing, 3-layer insulating glazing, argon-filled cavities described for standard panes	Public product documentation describes 3-layer glazing and insulating cavities with argon
Main system	Approximate dimensions	1230 x 1480 mm reference window	The product-specific EPD uses this reference size
Main system	Hardware	Top-guided opening hardware; exact hinge, friction arm and locking specification not fully detailed in the public summary	Use as a documented information gap unless a detailed technical drawing is added later
Main system	Gaskets and seals	Weather seals and insulating-glazing edge seals are expected, but exact gasket and seal specification is not fully confirmed in the public summary	Do not overclaim; treat exact seal material as not publicly confirmed
Main system	Fastening method	Installation method is documented through VELFAC installation guides and system drawings, but exact project-specific fixings are not defined for this generic case	Requires installation drawings for exact anchors, brackets, sealants and interface details
Main system	Surface treatment or cladding	Exterior aluminium cladding; interior timber with factory finish and water-based timber surface treatment	VELFAC EPD mentions factory-finished units and timber surface treatment
Repairability	Component access	Accessible for ordinary cleaning, operation, lubrication and adjustment; deeper component-level access requires technical guide confirmation	Based on public user, adjustment and maintenance guidance
Repairability	Replaceable elements	Fittings and accessories appear to have replacement guidance; likely serviceable elements include selected hardware, handles and adjustment-related parts	Public guides include replacement guides for fittings and accessories; exact complete list should not be assumed
Repairability	Non-replaceable or difficult elements	Insulated glazing unit edge seal, sealed glass assembly, composite frame junctions and factory-finished elements may be difficult to repair independently	Analytical assessment based on typical IGU and composite window construction; not directly confirmed as a VELFAC limitation
Repairability	Likely first failure points	Hardware wear or maladjustment, hinge and handle lubrication needs, gasket or seal ageing, glazing unit seal failure, surface or finish ageing, drainage or weather-exposure details	Derived from window literature and typical window maintenance logic; use as likely failure points, not measured failures
Repairability	Maintenance guidance	User, operation, maintenance, installation, replacement and adjustment guides are publicly listed by VELFAC	Useful enabling condition for repair and maintenance

Repairability	Spare-part availability	Spare parts can be ordered through VELFAC customer service, but long-term compatible spare-part availability is not clearly specified as a time guarantee	Treat as partial evidence and information gap
Repairability	Warranty	VELFAC warranty page states a 12-year warranty for VELFAC 200 window systems; glass warranty terms and other accessory periods vary	Use cautiously and refer to official warranty document for exact conditions
Repairability	Expected service life	Manufacturer warranty page states a 40-50 year design life	Use as manufacturer claim, not independently verified service-life evidence
Disassembly and reuse	Disassembly method	Not fully publicly confirmed for reuse-oriented disassembly; installation and system drawings may support removal, but circular disassembly instructions are not clearly provided	Important information gap for reuse and end-of-life analysis
Disassembly and reuse	Disassembly barriers	Potential barriers include sealed insulated glazing unit, composite material junctions, factory-finished assemblies, sealants, inaccessible fixings and lack of explicit reuse-oriented disassembly instructions	Analytical assessment; should be framed as possible barriers
Disassembly and reuse	Potential reuse route	Primary pathway is maintenance and selected component replacement during use; end-of-life pathway likely material recycling; direct reuse is uncertain because units are made to measure	Direct reuse depends on dimensions, removal condition, certification and local reuse market
Environmental evidence	Environmental product declaration available?	Yes	The product-specific EPD is available for VELFAC 200 ENERGY - Top-guided window
Environmental evidence	Product data sheet available?	Yes	Product information, system drawings and product database resources are available
Environmental evidence	Maintenance or installation manual available?	Yes	User, operation, maintenance, installation, replacement and adjustment resources are publicly listed
Environmental evidence	Main environmental uncertainty	The EPD excludes B1-B7 use phase; repair frequency, spare-part duration, actual service life, maintenance behaviour, end-of-life recovery and reuse feasibility remain uncertain	This is highly relevant for the limitations section
Relevance to study	Why this case was selected	The system combines a long-life wood-aluminium frame concept, triple glazing, aluminium exterior protection, public environmental documentation and partial maintenance or replacement guidance, making it suitable for analysing repairability, component replacement and lifecycle uncertainty	The case is strong because it has useful evidence but still exposes important missing information
Relevance to study	Link to research question	The case helps test whether a high-performance residential window system can support product-life extension, and where expected environmental benefits may be offset by missing repair information, spare-part uncertainty, material complexity, replacement timing, incomplete use-phase modelling or end-of-life uncertainty	Directly supports the research question on design and system conditions for repair and component replacement without offsetting environmental benefit